

Kepler Cross-staff Prototype 0.1 Build Guide

> ****Status:**** Prototype 0.1

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> This guide describes the construction of the first experimental Kepler Cross-staff. The purpose of this prototype is to evaluate the design, identify shortcomings, and guide future revisions. It is expected that both the instrument and this document will evolve.

Goal

Construct a simple, inexpensive cross-staff that can be used to make the first angular measurements of the night sky.

The objective is ****not**** maximum precision.

The objective is to create a scientifically useful prototype that can be evaluated through field use.

Materials

- 1 wooden yardstick
- 1 wooden paint stir stick
- 2 medium binder clips
- Pencil
- Tape
- Scissors

Optional:

- Printed labels
- Notebook for recording observations during construction and testing

Before You Begin

This is a prototype.

Do not worry about making it beautiful.

Instead, pay close attention to:

- what feels awkward;
- what feels intuitive;
- what is difficult to align;
- what seems unnecessarily complicated.

Your observations are as valuable as the instrument itself.

Step 1 – Prepare the Main Staff

Select one face of the yardstick to become the measurement face.

Keep the factory markings visible.

Do not modify the yardstick at this stage.

Choose one end to be the observing end and mark the reverse side:

``` text  
EYE  
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Always observe from this end.

Step 2 – Prepare the Crosspiece

The paint stir stick will become the movable crosspiece.

Do not cut or shorten it.

Find its midpoint and mark it with a pencil.

This mark should align with the centerline of the yardstick whenever the crosspiece is mounted.

Step 3 – Attach the Crosspiece

Place the paint stir stick across the yardstick at approximately a right angle.

Use one or two binder clips to hold it against the yardstick.

The crosspiece should:

- slide with moderate force;
- remain in position when released;
- remain approximately perpendicular to the yardstick.

Do not worry if this mechanism feels imperfect.

The purpose of Prototype 0.1 is to evaluate exactly these questions.

Step 4 – Initial Inspection

Hold the completed instrument in your hands.

Check the following:

- Does the crosspiece remain square to the staff?
- Does it slide smoothly?
- Does it stay where you place it?
- Does anything wobble?
- Does the instrument feel comfortable to hold?

Record your observations.

Step 5 – First Alignment Test

Indoors, choose two objects separated by a noticeable angle.

Examples include:

- opposite corners of a room;
- two pictures on a wall;
- a door frame and a window.

Hold the observing end near one eye.

Attempt to align:

- one end of the crosspiece with the first object;
- the opposite end of the crosspiece with the second object.

Repeat several times.

Do not attempt to measure anything yet.

The goal is simply to become familiar with the mechanics of the instrument.

Step 6 – Engineering Notes

While using the instrument, keep a notebook nearby.

Every time something surprises you, record it.

Examples:

- The binder clips interfere with my hand.
- I naturally close one eye.
- The crosspiece twists.
- The instrument is difficult to hold steady.
- I wanted a handle.
- I wasn't sure which edge to align.
- The yardstick flexes more than expected.

There are no incorrect observations.

Every difficulty is potential evidence that the design can be improved.

What Not to Do

Do not attempt to improve the design yet.

Use the instrument first.

Experience should guide design changes rather than assumptions.

Looking Ahead

After completing Prototype 0.1, the next steps will be:

1. Evaluate the instrument outdoors.
2. Develop a calibration procedure.
3. Compare the prototype with historical cross-staff designs.
4. Revise the design based on field experience.
5. Produce Prototype 0.2.

The Kepler Cross-staff is intended to evolve through observation, experimentation, and revision—the same scientific process it is designed to support.